

Harmonic and Computational Analysis of Odd Numbers via Truncated Dirichlet Series

The Strength of Mathematical Base Parameters ($b_{\text{mathematical}}$) vs. Reference Parameters ($b_{\text{reference}}$)

The strength of the developments in harmonic_analysis_odd_numbers_EN (vf1) lies in how the mathematical base parameters ($b_{\text{mathematical}}$) allow for before-and-after products that are consistently structured to indicate primality, unlike reference parameters ($b_{\text{reference}}$).

1. Testing with Positive Zeta Zeros ($b = \gamma_n$)

$b (= \gamma_n)$	$\text{Re}(S_{f4})$	$\text{Re}(S_{f6})$	$\text{Re}(\text{product})$	Sign
14.134725142	-0.859226	-0.826011	+0.723770	> 0
21.022039639	-0.880357	-0.149303	+0.160232	> 0
25.010857580	-0.859549	-0.087674	+0.078383	> 0
30.424876126	-0.799171	-0.866032	+0.270869	> 0

2. Testing with Negative Zeta Zeros ($b = -\gamma_n$)

$b (= -\gamma_n)$	$\text{Re}(S_{f4})$	$\text{Re}(S_{f6})$	$\text{Re}(\text{product})$	Sign
-14.134725142	-0.859226	-0.826011	+0.723770	> 0
-21.022039639	-0.880357	-0.149303	+0.160232	> 0
-25.010857580	-0.859549	-0.087674	+0.078383	> 0
-30.424876126	-0.799171	-0.866032	+0.270869	> 0

3. Comprehensive Analysis Across 500 Zeros

- Global Distribution:** Positives: 267 (53.4%); Negatives: 233 (46.6%); Z-score vs 50/50: 1.52.
- Run Structure:** 280 runs observed, max length = 5. Slight tendency to alternation.
- Correlations:** No robust dependency on primality or parity of n .